

Bijal Panchal

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SUMMARY

Data Analytics Engineer & BI Professional (M.S. Data Analytics with Distinction). I design star schema warehouses consolidating millions of records, build automated Python ETL pipelines, and cut data inconsistencies by 30%. Delivered 10+ Tableau/Power BI dashboards adopted for executive decision-making. Skilled in SQL, Python, and dimensional modeling, with strong client-facing communication translating complex data into actionable business insights. Currently expanding expertise in Snowflake, dbt, and GCP. Available immediately for full-time analytics roles.

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn), SQL, VBA, SQLAlchemy

BI/Visualization: Tableau, Power BI (DAX, Power Query), Streamlit, Excel (VBA-automated), PowerPoint

Modeling: K-Means Clustering, XGBoost Classification, PCA, RFM Segmentation, ARIMA, SARIMA, Holt-Winters, Prophet, Z-Score Anomaly Detection, Isolation Forest, Feature Engineering, Model Evaluation (Precision, Recall, F1-Score, AUC-ROC)

Warehousing: MySQL, SQLite, PostgreSQL, Star Schema Design, ETL Pipelines, Dimensional Modelling, Data Warehousing

DataOps: Data Standardization, Completeness/Accuracy/Consistency Checks, Duplicate & Missing-Value Resolution, Data Quality Management, Reconciliation, Root Cause Analysis

DevTools: Jupyter Notebook, VS Code, Git, GitHub, GitHub Actions, DBeaver, pytest

CloudStack: Snowflake, dbt, GCP BigQuery, AWS S3/Redshift, Docker

Methodologies: Analytics Engineering, KPI Framework Development, Customer Segmentation, Executive Reporting & Stakeholder Communication, Cross-functional Collaboration, Agile, SDLC

WORK EXPERIENCE

POINTS AFRICA

AUG 2025 - JAN 2026

Data Analyst Intern

Toronto, ON (Remote)

- Unified and standardized fragmented datasets from **3 enterprise partners** (GLICO, JUMIA, STAR) spanning customer profiles, transactions, loyalty ledgers, and geographic data — cutting data inconsistencies by **30%** via automated Python (Pandas, NumPy) and SQL pipelines.
- Achieved **100% transactional field coverage** by building iterative ETL workflows that resolved phone formatting errors, date-time mismatches, and duplicate records at scale.
- Architected a **star schema data model** (4 dimension tables + central Transaction Fact table), consolidating multi-partner data into a single source of truth and boosting query performance by **40%**.
- Applied **RFM segmentation** to classify customer loyalty tiers, surfacing high-value clusters across 3 transaction networks to drive targeted partner engagement strategies.
- Delivered **10+ production-ready Tableau dashboards** — covering KPIs, partner performance, payment behavior, and geo-regional trends — directly adopted for internal decision-making.
- Translated complex, multi-source data into clear executive-level narratives, enabling senior stakeholders to make confident, data-driven decisions without needing direct database access.

DMG GROUP OF COMPANIES

May 2024 - Dec 2024

Data Analytics & ML Intern

Ahmedabad

- Built a **customer segmentation model** for IT service clients using Python (Scikit-learn, Pandas), combining K-Means Clustering and XGBoost classification to achieve **87% classification accuracy** in identifying high-value client segments.
- Engineered **12+ behavioral and transactional features** from client usage and contract data, reducing data noise by **28%** through advanced feature selection and PCA-based dimensionality reduction.
- Developed interactive client performance and revenue dashboards in **Excel (VBA-automated)** and **Power BI**, consolidating data across **4+ service verticals** for real-time tracking of acquisition, utilization, and revenue trends.
- Cleaned and standardized **4,000+ client records** across service and billing datasets using Python and MySQL ETL workflows, improving overall data quality by **30%**.
- Evaluated model performance (Precision, Recall, F1-Score, AUC-ROC) and presented actionable insights to management — directly supporting decisions that improved client retention by **18%**.

PROJECTS

Global City Traffic Intelligence Platform | Python, SQLite, Streamlit, Scikit-learn, Prophet — Live

- Engineered a **fully automated real-time data platform** ingesting daily metrics across **5 global cities** from **5 live APIs** (TomTom, OpenAQ, Open-Meteo, Numbeo, City Open Data) — built on modular Python collectors feeding a SQLite star schema warehouse (**9 tables, 4,400+ rows**) via a self-running GitHub Actions pipeline.
- Designed a **weighted City Stress Index scoring engine** (Traffic 35% + Air Quality 25% + Weather 20% + Safety 10% + Cost of Living 10%) powered by **15+ engineered features** — including rolling averages, volatility scores, and heat stress index — using ARIMA, Prophet, Z-Score, Isolation Forest, and K-Means models.
- Deployed a **7-page interactive Streamlit dashboard** (live at city-stress-index.streamlit.app) with real-time stress scores, 30-day forecasts, anomaly detection alerts, and cross-city clustering — validated by a **22-test pytest suite** ensuring pipeline and database integrity.

Retail Sales Forecasting & Demand Planning | Python, SQL, Power BI, Statsmodels, SQLite

- Cleaned and processed **1M+ transactional records** across **43 countries and 4,735 products**, resolving 34K duplicates, 19K cancellations, and 243K missing customer IDs — then engineered a SQL star schema warehouse (1 fact + 4 dimension tables) via a custom Python ETL pipeline (SQLAlchemy).
- Built and benchmarked **SARIMA and Holt-Winters** forecasting models on monthly revenue data, with SARIMA achieving **74.5% accuracy** (MAPE 25.54%) on a chronological 80/20 train-test split — reducing simulated stockout rates by **18%**.
- Delivered a **4-page Power BI dashboard** with dynamic KPI cards, seasonal trend lines, and forecast-vs-actual comparisons (80% confidence intervals) — powering RFM segmentation across 6 customer tiers that identified **843 Champions** (avg. \$5,689 spend) and flagged 409 At-Risk customers as the top re-engagement priority.

EDUCATION

University of Niagara Falls

OCT 2024 - APR 2026

Data Analytics

- Graduated with Distinction, GPA 3.89/4.0 — President's Honour Roll, 100% in Advanced Analytics Internship.

Navrachana University

JUL 2020 - JUN 2024

Computer Science & Engineering

- Coursework in Data Structures, DBMS, AI, Cloud Computing, and Cybersecurity; completed a 12-credit Industry Internship and Major Project.

CERTIFICATIONS & ACTIVITIES

Advanced Power BI: DAX Language, Formulas, and Calculations | Course Completion – Apr 2025

- A targeted Power BI certification covering core skills in data modeling, DAX language, and advanced DAX formulas. Emphasizes practical application through dynamic dashboards and real-world business intelligence scenarios.

Hack For Change | Participant – Apr 2022

- Participated in a 49-hour hackathon organized by Navrachana University in collaboration with NuvYuva. Collaborated with peers to develop innovative solutions aimed at social impact and technology-driven change. Strengthened skills in project development, teamwork, and rapid problem-solving under time constraints.